CS520: Analysis of algorithms

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Local occupancy after a block update

Our starting point

Fix a graph G on n vertices and $\lambda > 0$.

- Throughout this lecture, start either dynamics from $X_0 = \emptyset$
- The empty set is independent in every graph, and every update preserves independence
- Thus $\mu_0 = \delta_\emptyset$ and for the transition kernel K the law at time t is

$$\mu_t = \delta_\emptyset K^t$$

- We will apply the local inequalities to this changing distribution at each step

Conditioning works from every current state

Fix a current state I and a vertex u. Form the available graph F using the exterior $S = I \setminus N(u)$. Now consider applying the kernel $Q_u P_u$.

- After Q_u:
  - the probability that the neighborhood is all unoccupied is $1/Z_F(\lambda)$
  - the expected number of occupied vertices is $E_F(\lambda) = \lambda Z_F'(\lambda)/Z_F(\lambda)$
- The subsequent P_u update occupies u with probability p precisely when that neighborhood is empty
- Recall that δ_I is the point mass with 1 at I and zeros elsewhere

Local occupancy gives…

- Then

$$\mathbb{E}_{\delta_I Q_u P_u} x_u = \frac{p}{Z_F(\lambda)}, \qquad \mathbb{E}_{\delta_I Q_u} y_u = E_F(\lambda)$$

- And since we have the local occupancy inequality for every possible F,

$$1 \leq \beta \mathbb{E}_{\delta_I Q_u P_u} x_u + \gamma \mathbb{E}_{\delta_I Q_u} y_u$$

- If μ is an arbitrary distribution over independent sets then $\mu = \sum_I \mu_I \delta_I$ and hence by averaging we get the **block inequality (B)**:

$$1 \leq \beta \mathbb{E}_{\mu Q_u P_u} x_u + \gamma \mathbb{E}_{\mu Q_u} y_u.$$

The extra power for algorithms



- This is why we consider **every induced neighborhood subgraph** for local occupancy
- Crucially, we **do not assume** that the current distribution is the hard-core distribution
- The current distribution can be whatever our algorithm has got to so far
- If the updates barely change the current distribution μ, then (B) should nearly hold with μ in both expectations instead of $\mu Q_u P_u$ and μQ_u
- Summing those approximate inequalities will force $\mathbb{E}_\mu |X|$ to be large even though μ is **not** exactly the hard-core model

The counting identity again

For every independent set I,

$$\sum_u x_u(I) = |I|, \qquad \sum_u y_u(I) = \sum_{v \in I} d(v) \leq \Delta |I|.$$

Consequently, for **any** distribution μ,

$$\sum_u \left(\beta \mathbb{E}_\mu x_u + \gamma \mathbb{E}_\mu y_u \right) \leq \Gamma \mathbb{E}_\mu |X|.$$

We now need a quantitative meaning of “barely change.”

The proof intuition

If local updates barely change μ



- **Suppose** $\mu Q_u \approx \mu$ and $\mu Q_u P_u \approx \mu$: resampling barely changes the distribution
- The expectations in (B) should then barely change, giving

$$1 - \epsilon \leq \beta \mathbb{E}_\mu x_u + \gamma \mathbb{E}_\mu y_u$$

- Summing over u would give $\mathbb{E}_\mu |X| \geq (1 - \epsilon) n / \Gamma$
- Thus, it suffices to know the update changes are small **on average over locations**.

Why should local changes become small quickly?



- A heat-bath update replaces one region's conditional law by the correct one under π: it removes a local discrepancy between μ and π
- Other updates may disturb that region again so we track a **potential** quantifying the remaining discrepancy
- A suitable potential will:
  - Start at $O(n)$ from the empty set, for fixed λ
  - Decrease as the algorithm progresses
  - Decrease quickly if the local update changes μ a lot
- We show that **relative entropy** has all these properties

Comparing probability distributions

Total variation distance

For distributions μ, ρ on a finite space, define

$$d_{\mathrm{TV}}(\mu, \rho) = \frac{1}{2} \sum_{I} |\mu(I) - \rho(I)|.$$

- The key to working with TV is to split the sum into positive and negative terms
- Because $\sum_I (\mu(I) - \rho(I)) = 0$, the positive and negative sums have the same absolute value
- Taking $A = \{I : \mu(I) > \rho(I)\}$ therefore gives

$$d_{\mathrm{TV}}(\mu, \rho) = \mu(A) - \rho(A) = \max_B |\mu(B) - \rho(B)|$$

- This distance lies in $[0, 1]$ and satisfies the triangle inequality

Total variation controls expectations

If $0 \leq f \leq M$, then

$$|\mathbb{E}_\mu f - \mathbb{E}_\rho f| \leq M d_{\mathrm{TV}}(\mu, \rho).$$

- How can we prove this using the previous slide?
- We will apply this with $x_u \in [0, 1]$ and $y_u \in [0, d(u)]$

Total variation distance and events



Lemma: Total variation distance is the maximum difference in probability the two measures can give to the same event.

Proof: Will be a homework exercise.

Lemma: applying a kernel cannot increase TV distance

For every kernel K,

$$d_{\mathrm{TV}}(\mu K, \rho K) \le d_{\mathrm{TV}}(\mu, \rho).$$

Proof: use the triangle inequality, nonnegative entries, and row sums:

$$\begin{aligned} 2d_{\mathrm{TV}}(\mu K, \rho K) &= \sum_J \left| \sum_I (\mu(I) - \rho(I)) K(I,J) \right| \\ &\le \sum_I |\mu(I) - \rho(I)| \sum_J K(I,J) \\ &= 2d_{\mathrm{TV}}(\mu, \rho). \end{aligned}$$

Relative entropy as a potential

Reminder: entropy measures uncertainty

The Shannon entropy of a distribution μ is

$$\mathrm{Ent}(\mu) = -\sum_{I} \mu(I) \log \mu(I),$$

with $0 \log 0 = 0$.

- A certain outcome has entropy 0
- A uniform choice among m outcomes has entropy $\log m$
- The quantity $-\log \mu(I)$ measures the surprise of outcome I; entropy is its average
- Our potential will compare μ with the generally nonuniform distribution π

Relative entropy

For distributions μ, ρ, define the **relative entropy** or **KL divergence**

$$D(\mu \parallel \rho) = \sum_{I:\mu(I)>0} \mu(I) \log \frac{\mu(I)}{\rho(I)}.$$

- If $\rho(I) = 0 < \mu(I)$, the divergence is $+\infty$
- It is not *technically* a distance: it need not be symmetric
- It is true that $D(\mu \parallel \rho) \geq 0$ with equality iff $\mu = \rho$, which we prove soon

Relative entropy as extra surprise

- Suppose outcomes come from μ, but we assign probabilities using ρ
- We then assign surprise $-\log \rho(I)$ instead of $-\log \mu(I)$. The average *extra* surprise is

$$\mathbb{E}_{I\sim\mu}[-\log \rho(I)] - \mathrm{Ent}(\mu) = D(\mu \parallel \rho)$$

- If ρ is uniform on m states, this is $\log m - \mathrm{Ent}(\mu)$: the gap from maximum entropy
- Taking $\rho = \pi$ extends this comparison to our **nonuniform** equilibrium law

Our potential is relative entropy to hard-core



- For any distribution μ on the independent sets, write $\Phi(\mu) \coloneqq D(\mu \parallel \pi)$
- Let $h(x) = x \log x$, with $h(0) = 0$. Then

$$\Phi(\mu) = \sum_I \mu(I) \log\left(\frac{\mu(I)}{\pi(I)}\right) = \sum_I \pi(I) h\left(\frac{\mu(I)}{\pi(I)}\right)$$

- The hard-core distribution is positive on every state, so our potential $\Phi(\mu)$ is always finite
- We will prove that updates to μ decrease $\Phi(\mu)$ by an amount that controls the effects of the updates
- Φ will help us prove that our algorithms do something useful

Why relative entropy is nonnegative

- $h(x) = x \log x$ is strictly convex on $[0, \infty)$, since $h''(x) = 1/x > 0$ for $x > 0$
- If $\mu(I) > 0 = \rho(I)$ for some I, then $D(\mu \parallel \rho) = +\infty$
- Otherwise, Jensen's inequality with weights $\rho(I)$ (which sum to 1) gives

$$D(\mu \parallel \rho) = \sum_{I:\rho(I)>0} \rho(I) h\left(\frac{\mu(I)}{\rho(I)}\right) \geq h\left(\sum_{I:\rho(I)>0} \rho(I) \frac{\mu(I)}{\rho(I)}\right) = h(1) = 0.$$

- By strict convexity, equality holds exactly when all ratios $\mu(I)/\rho(I)$ are 1, i.e. $\mu = \rho$
- In particular, our potential Φ can never fall below 0

Convexity of relative entropy (C)

- If $\mu = \sum_j w_j \mu_j$, where $w_j \geq 0$ and $\sum_j w_j = 1$, then

$$\Phi(\mu) \leq \sum_j w_j \Phi(\mu_j)$$

Proof: Recall that $h(x) = x \log x$, with $h(0) = 0$, is convex. Apply Jensen with weights w_j at each fixed state I, then sum with weights $\pi(I)$:

$$\Phi(\mu) = \sum_I \pi(I) h\left(\frac{\mu(I)}{\pi(I)}\right) = \sum_I \pi(I) h\left(\sum_j w_j \frac{\mu_j(I)}{\pi(I)}\right)$$

$$\leq \sum_I \pi(I) \sum_j w_j h\left(\frac{\mu_j(I)}{\pi(I)}\right) = \sum_j w_j \Phi(\mu_j)$$

Convexity of relative entropy (C)



- This means that mixing distributions cannot increase their average potential
- With this setup, to sample from μ you first choose an index j with probability w_j, then draw a state from μ_j
- This is how we will analyze the random choice of update location

Pinsker's inequality (P)

We need the inequality

$$D(\mu \parallel \rho) \geq 2d_{\mathrm{TV}}(\mu, \rho)^2.$$

- We will use it to converts an entropy decrease into a bound on how much an update changes the distribution in TV distance
- The constant 2 corresponds to studying entropy with natural logarithms
- Here is a proof using only convexity and one-variable calculus

First, combine outcomes using Jensen's inequality

For $a_i \geq 0$ and $b_i > 0$, write $A = \sum_i a_i$ and $B = \sum_i b_i$.

Recall that $h(x) = x \log x$ is convex. **Jensen's inequality** with weights b_i / B gives

$$\sum_i a_i \log \frac{a_i}{b_i} = B \sum_i \frac{b_i}{B} h\left(\frac{a_i}{b_i}\right)$$
$$\geq B h(A/B) = A \log(A/B).$$

Terms with $a_i = b_i = 0$ may be omitted; if $a_i > 0 = b_i$, the left side is infinite. When $A = 0$, both sides are 0.

Pinsker reduces to two outcomes

Take $B = \{I : \mu(I) > \rho(I)\}$, $r = \mu(B)$ and $q = \rho(B)$, so $r - q = d_{\mathrm{TV}}(\mu, \rho)$.

Jensen on B and its complement, with weights proportional to $\rho(I)$, gives

$$D(\mu \parallel \rho) \geq r \log \frac{r}{q} + (1 - r) \log \frac{1 - r}{1 - q} =: g_q(r).$$

For fixed $0 < q < 1$ and $0 < r < 1$, differentiation gives

$$g_q'(r) = \log \frac{r}{q} - \log \frac{1 - r}{1 - q},$$

$$g_q''(r) = \frac{1}{r} + \frac{1}{1 - r} = \frac{1}{r(1 - r)} \geq 4.$$

Integrating twice: slope, then value

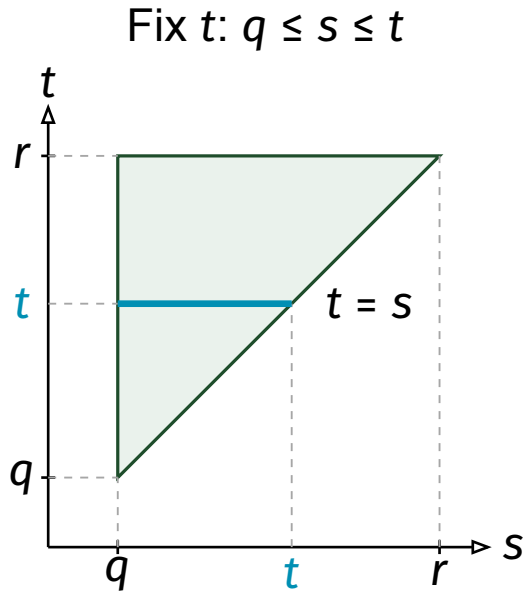
- Assume $0 < q \leq r < 1$, fix q, and note that $g_q(q) = g'_q(q) = 0$ by substitution
- Integrating first recovers the slope, then the value:

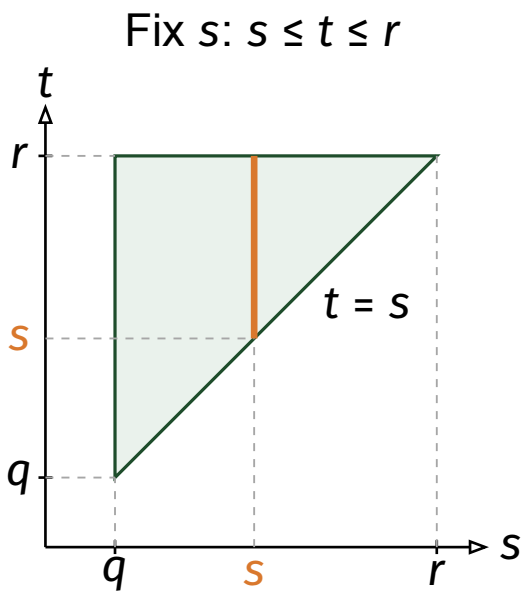
$$g'_q(t) = g'_q(q) + \int_q^t g''_q(s)\,ds = \int_q^t g''_q(s)\,ds,$$

$$g_q(r) = g_q(q) + \int_q^r g'_q(t)\,dt = \int_q^r \int_q^t g''_q(s)\,ds\,dt.$$

- The region is $s, t \in \mathbb{R}$ such that $q \leq s \leq t \leq r$
- If we interchange the order of integration, the inner integrand is a constant wrt t so we just get the length $(r - s)$ of the integral…

Illustration: reversing the integration order





$$\int_q^r \int_q^t g_q''(s)\,ds\,dt = \int_q^r \int_s^r g_q''(s)\,dt\,ds = \int_q^r (r-s) g_q''(s)\,ds$$

Finishing Pinsker's inequality

Now use $g_q''(s) \geq 4$ and $r - s \geq 0$:

$$g_q(r) = \int_q^r (r-s) g_q''(s)\, ds \geq 4 \int_q^r (r-s)\, ds = 2(r-q)^2.$$

Thus

$$D(\mu \parallel \rho) \geq 2(r-q)^2 = 2d_{\mathrm{TV}}(\mu,\rho)^2. \qquad \blacksquare$$

- Values $r = 0, 1$ follow by continuity
- If $q = 0$ or $q = 1$, either the distributions agree on this event or the relevant divergence is infinite, so the inequality still holds

The chain rule for entropy

- Reveal a pair $(X, Y) \sim \mu$ in stages: first X, then Y
- μ_X is the distribution of X; $\mu(\cdot \mid x)$ is the distribution of Y given $X = x$
- Probabilities multiply: $\mu(x, y) = \mu_X(x)\mu(y \mid x)$, so surprises (logs of probabilites) add. Averaging gives the **chain rule for entropy**:

$$\mathrm{Ent}(\mu) = \mathrm{Ent}(\mu_X) + \sum_{x:\mu_X(x)>0} \mu_X(x)\mathrm{Ent}(\mu(\cdot \mid x)).$$

- Total uncertainty = first-stage uncertainty + average remaining uncertainty

The chain rule for relative entropy

For two joint distributions μ, ρ with $D(\mu \parallel \rho) < \infty$, split the average **extra surprise** between the same two reveals:

$$D(\mu \parallel \rho) = D(\mu_X \parallel \rho_X) + \sum_{x:\mu_X(x)>0} \mu_X(x) D(\mu(\cdot \mid x) \parallel \rho(\cdot \mid x)).$$

- First term: disagreement about the first reveal, X
- Second term: remaining disagreement about Y given X, averaged with weights $\mu_X(x)$ because outcomes follow μ
- X and Y need not be independent
- In our application, take X to be the exterior configuration and Y the interior configuration

The chain rule: reveal exterior, then interior



- Let $E \subset V$ be a set of "exterior" vertices, and write a state I as (s, a), where s is state on E and a is state on $V \setminus E$
- Think of I as a vector in $\{0,1\}^V$ with $I_v = 1$ meaning $v \in I$
- List the coordinates with E first, then $V \setminus E$
- Let μ_E be the exterior marginal and $\mu(\cdot \mid s)$ the conditional interior law
  - μ_E is the distribution on $\{0,1\}^E$ you get by generating I with μ and ignoring the coordinates in $V \setminus E$
  - $\mu(\cdot \mid s)$ is the probability distribution on $\{0,1\}^{V \setminus E}$ you get by fixing s in the exterior and sampling the interior a from μ conditioned on the exterior s

The chain rule: reveal exterior, then interior

- Equivalently, you get $\mu(\cdot, s)$ by generating I with μ repeatedly until the exterior coordinates agree with s and then ignoring the exterior coordinates

Conditional probability

- Conditional probability means:

$$\mu(s,a) = \mu_E(s)\mu(a \mid s), \quad \pi(s,a) = \pi_E(s)\pi(a \mid s)$$

- Hence

$$\log \frac{\mu(s,a)}{\pi(s,a)} = \log \frac{\mu_E(s)}{\pi_E(s)} + \log \frac{\mu(a \mid s)}{\pi(a \mid s)}$$

The resulting entropy decomposition

- We can now prove that, for any exterior E we have

$$\Phi(\mu) = D(\mu \parallel \pi) = \sum_{(s,a)} \mu(s,a) \log \frac{\mu(s,a)}{\pi(s,a)}$$

$$= \sum_{(s,a)} \mu_E(s)\mu(a \mid s)\left(\log \frac{\mu_E(s)}{\pi_E(s)} + \log \frac{\mu(a \mid s)}{\pi(a \mid s)}\right)$$

$$= \sum_{s} \mu_E(s) \log \frac{\mu_E(s)}{\pi_E(s)} + \sum_{s} \mu_E(s) \sum_{a} \mu(a \mid s) \log \frac{\mu(a \mid s)}{\pi(a \mid s)}$$

$$= D(\mu_E \parallel \pi_E) + \sum_{s} \mu_E(s) D(\mu(\cdot \mid s) \parallel \pi(\cdot \mid s)).$$

The math meets the algorithm

$$\Phi(\mu) = D(\mu_E \parallel \pi_E) + \sum_s \mu_E(s) D(\mu(\cdot \mid s) \parallel \pi(\cdot \mid s)).$$

- The first term measures disagreement on the exterior
- The second measures average conditional disagreement inside (averaged over the probabilities μ_E gives to independent sets in the exterior)
- Exteriors with $\mu_E(s) = 0$ contribute zero and can be omitted
- A heat-bath update to the interior zeroes out the second term and leaves the first unchanged

What the heat-bath update removes

- Let kernel R keep the exterior s fixed and resample the interior from $\pi(\cdot \mid s)$
- The updated law satisfies

$$(\mu R)(s,a) = \mu_E(s)\pi(a \mid s), \qquad (\mu R)_E = \mu_E.$$

- Compare the updated law μR with π using the chain rule:

$$\Phi(\mu R) = D(\mu_E \,\|\, \pi_E) + \sum_s \mu_E(s) \underbrace{D(\pi(\cdot \mid s) \,\|\, \pi(\cdot \mid s))}_{0}$$

$$= D(\mu_E \,\|\, \pi_E)$$

The heat-bath entropy identity (H)

- Now compare the original law μ with the updated law μR
- Their exterior marginals agree, so the chain rule gives

$$D(\mu \,\|\, \mu R) = \underbrace{D(\mu_E \,\|\, \mu_E)}_{0} + \sum_s \mu_E(s) D(\mu(\cdot \mid s) \,\|\, \pi(\cdot \mid s))$$

$$= \Phi(\mu) - \Phi(\mu R)$$

- The sum is exactly the term zeroed out of $\Phi(\mu)$ on the previous slide
- Applying Pinsker gives

$$\Phi(\mu) - \Phi(\mu R) = D(\mu \,\|\, \mu R) \geq 2 d_{\mathrm{TV}}(\mu, \mu R)^2$$

- So a large change in distribution (in TV) requires a large potential drop

Why the identity applies here



- For P_u, take the interior to be $\{u\}$
- For Q_u, take the interior to be $N(u)$
- In both cases the exterior marginal stays fixed and the interior becomes its conditional hard-core distribution
- Thus (H) applies to either update, from **every** distribution μ
- Intuition: a heat-bath update corrects a conditional distribution; a large correction costs a large amount of relative entropy

The heart of the algorithm analysis

Initial potential: only the empty set contributes

- Our initial law $\mu_0 = \delta_\varnothing$ assigns probability 1 to $\varnothing$ and 0 to every other state
- Since $|\varnothing| = 0$, its hard-core probability is

$$\pi(\varnothing) = \frac{\lambda^0}{Z_G(\lambda)} = \frac{1}{Z_G(\lambda)}$$

- The relative-entropy sum has just one nonzero term:

$$\Phi_0 := \Phi(\mu_0) = 1 \cdot \log \frac{1}{\pi(\varnothing)} = \log Z_G(\lambda)$$

Bounding the initial potential

Summing over **all subsets**, including those that are not independent, gives

$$Z_G(\lambda) = \sum_{I \in \mathcal{I}(G)} \lambda^{|I|} \leq \sum_{S \subseteq V} \lambda^{|S|} = (1 + \lambda)^n.$$

Each vertex contributes a factor 1 when absent or λ when present.

Taking logarithms gives our initial entropy budget:

$$\Phi_0 = \log Z_G(\lambda) \leq n \log(1 + \lambda).$$

- This is a computable upper bound: we do not need to calculate Z_G
- For fixed λ, the budget is $O(n)$; at $\lambda = 1$, it is at most $n \log 2$

Entropy decreases telescope over time

For either dynamics, define the one-step decrease $D_K(\mu) = \Phi(\mu) - \Phi(\mu K)$. Since $\mu_t = \delta_\varnothing K^t$, we have $\mu_t K = \mu_{t+1}$.

For any integer $T \geq 1$, sum the decreases and shift the second sum's index:

$$\begin{aligned}\sum_{t=0}^{T-1} D_K(\mu_t) &= \sum_{t=0}^{T-1} \left(\Phi(\mu_t) - \Phi(\mu_{t+1})\right) \\ &= \sum_{t=0}^{T-1} \Phi(\mu_t) - \sum_{t=1}^{T} \Phi(\mu_t) \\ &= \Phi(\mu_0) - \Phi(\mu_T).\end{aligned}$$

The total decrease is bounded by the initial budget



- Relative entropy is nonnegative, so $\Phi(\mu_T) \geq 0$
- With $\Phi(\mu_0) = \Phi_0$, the telescoping identity gives

$$\sum_{t=0}^{T-1} D_K(\mu_t) = \Phi_0 - \Phi(\mu_T) \leq \Phi_0 \leq n \log(1 + \lambda).$$

- By convexity (C) and the heat-bath identity (H), each $D_K(\mu_t) \geq 0$: each step spends some of this fixed budget

Dividing by T, the **average decrease** is therefore small:

$$\frac{1}{T} \sum_{t=0}^{T-1} D_K(\mu_t) \leq \frac{\Phi_0}{T}.$$

Bounding an average of square roots using Jensen (CS)



- Let $z_0, \ldots, z_{T-1} \geq 0$, with $T \geq 1$.
- The function $f(x) = x^2$ is convex, so Jensen with equal weights $1/T$ at the points $\sqrt{z_t}$ gives

$$\left(\frac{1}{T}\sum_{t<T}\sqrt{z_t}\right)^2 \leq \frac{1}{T}\sum_{t<T}\left(\sqrt{z_t}\right)^2 = \frac{1}{T}\sum_{t<T} z_t.$$

- Taking square roots (the average on the left is nonnegative),

$$\frac{1}{T}\sum_{t<T}\sqrt{z_t} \leq \sqrt{\frac{1}{T}\sum_{t<T} z_t}.$$

- This is also a consequence of the **Cauchy–Schwarz** inequality

Averaging the entropy decreases

Apply (CS) with $z_t = D_K(\mu_t) \geq 0$. Our bound on the average entropy decrease gives

$$\frac{1}{T}\sum_{t<T}\sqrt{D_K(\mu_t)} \leq \sqrt{\frac{1}{T}\sum_{t<T} D_K(\mu_t)} \leq \sqrt{\frac{\Phi_0}{T}}.$$

Choosing a uniform time

Let τ be independent of the chain and uniform on $\{0, \ldots, T-1\}$.

- Each time has probability $1/T$
- Independence means conditioning on $\tau = t$ does not change the law of X_t
- The law of total expectation therefore gives

$$\mathbb{E}|X_\tau| = \frac{1}{T}\sum_{t<T} \mathbb{E}_{\mu_t}|X|.$$

This lets us express the average of the expected sizes as the expected size at one random time.

The neighborhood block theorem

Statement: local occupancy gives a finite-time bound



Theorem. Suppose local occupancy holds with parameters λ, β, γ, and put $\Gamma = \beta + \gamma\Delta$. For neighborhood block dynamics from $X_0 = \varnothing$ and integer $T \geq 1$,

$$\mathbb{E}|X_\tau| \geq \frac{n}{\Gamma}\left(1 - \sqrt{\frac{(\Gamma^2 + \beta^2)\Phi_0}{2T}}\right).$$

In particular, for $0 < \epsilon < 1$,

$$T \geq \frac{(\Gamma^2 + \beta^2)n \log(1+\lambda)}{2\epsilon^2} \quad \Rightarrow \quad \mathbb{E}|X_\tau| \geq (1-\epsilon)\frac{n}{\Gamma}.$$

Step 1: measure the two changes within a block

Fix any distribution μ and define

$$a_u = d_{\mathrm{TV}}(\mu, \mu Q_u), \qquad b_u = d_{\mathrm{TV}}(\mu Q_u, \mu Q_u P_u).$$

Convexity and two applications of (H) give

$$\begin{aligned} D_{KN}(\mu) &\geq \frac{1}{n} \sum_u \left(\Phi(\mu) - \Phi(\mu Q_u P_u) \right) \\ &= \frac{1}{n} \sum_u \left(\Phi(\mu) - \Phi(\mu Q_u) + \Phi(\mu Q_u) - \Phi(\mu Q_u P_u) \right) \\ &\geq \frac{2}{n} \sum_u (a_u^2 + b_u^2). \end{aligned}$$

Step 2: move the local inequality back to μ

The triangle inequality gives

$$d_{\mathrm{TV}}(\mu, \mu Q_u P_u) \leq a_u + b_u.$$

Since $0 \leq x_u \leq 1$ and $0 \leq y_u \leq d(u)$, (TV) gives

$$\mathbb{E}_{\mu Q_u P_u} x_u \leq \mathbb{E}_\mu x_u + a_u + b_u,$$

$$\mathbb{E}_{\mu Q_u} y_u \leq \mathbb{E}_\mu y_u + d(u) a_u.$$

Substitute these into the block inequality (B):

$$1 \leq \beta \mathbb{E}_\mu x_u + \gamma \mathbb{E}_\mu y_u + (\beta + \gamma d(u)) a_u + \beta b_u.$$

Step 3: sum over centers

Use the counting identity and $\beta + \gamma d(u) \leq \Gamma$:

$$1 \leq \frac{\Gamma}{n}\mathbb{E}_\mu |X| + \frac{1}{n}\sum_u (\Gamma a_u + \beta b_u).$$

Expanding $(\beta a - \Gamma b)^2 \geq 0$ gives $(\Gamma a + \beta b)^2 \leq (\Gamma^2 + \beta^2)(a^2 + b^2)$. Then apply (CS) with $z_u = a_u^2 + b_u^2$:

$$\frac{1}{n}\sum_u (\Gamma a_u + \beta b_u) \leq \sqrt{(\Gamma^2 + \beta^2)\frac{1}{n}\sum_u (a_u^2 + b_u^2)}$$

$$\leq \sqrt{\frac{(\Gamma^2 + \beta^2) D_{KN}(\mu)}{2}}.$$

Step 4: average over time

Apply the preceding inequality at μ_t and average:

$$1 \leq \frac{\Gamma}{n}\mathbb{E}|X_\tau| + \sqrt{\frac{\Gamma^2+\beta^2}{2}}\frac{1}{T}\sum_{t<T}\sqrt{D_{KN}(\mu_t)}$$

$$\leq \frac{\Gamma}{n}\mathbb{E}|X_\tau| + \sqrt{\frac{(\Gamma^2+\beta^2)\Phi_0}{2T}}.$$

Rearranging proves the theorem.

- The proof did not need a bound on the distance from μ_t to π
- It also did not require individual trajectories to increase in size

Transferring the argument to Glauber dynamics

Prescribed updates and the local certificate (G)



For each center u, choose a fixed list of vertices in its neighborhood,

$$(v_{u,1}, \ldots, v_{u,\ell_u}), \qquad S_u = P_{v_{u,1}} \cdots P_{v_{u,\ell_u}}.$$

- The list may be empty, and repetitions are allowed
- It is fixed independently of the starting state and of all random choices
- Suppose it achieves the following inequality from every input distribution μ:

$$\beta \mathbb{E}_{\mu S_u P_u} x_u + \gamma \mathbb{E}_{\mu S_u} y_u \geq 1.$$

- Exact simulation $S_u = Q_u$ suffices by (B), but (G) is all the proof needs

Count how often each vertex is used

Define

$$M_{uv} = \left|\{\, i : v_{u,i} = v \,\}\right|, \qquad \kappa = \max_v \sum_u M_{uv}, \qquad C = \beta + \kappa\Gamma.$$

- M_{uv} counts occurrences of v in the list for center u
- κ counts the most occurrences of any vertex across **all** lists
- If every list is empty, then $\kappa = 0$
- If each list visits every neighbor once, $M_{uv} = \mathbf{1}_{\{v \in N(u)\}}$ and $\kappa \le \Delta$
- These lists are used in the proof; the actual algorithm still chooses a fresh uniform vertex at each step

The Glauber transfer theorem

Theorem. If (G) holds for all u, μ, then ordinary Glauber dynamics started at $X_0 = \varnothing$ satisfies, for any integer $T \geq 1$,

$$\mathbb{E}|X_\tau| \geq \frac{n}{\Gamma}\left(1 - C\sqrt{\frac{\Phi_0}{2T}}\right), \qquad C = \beta + \kappa\Gamma.$$

In particular, for $0 < \epsilon < 1$,

$$T \geq \frac{C^2 n \log(1+\lambda)}{2\epsilon^2} \quad \Rightarrow \quad \mathbb{E}|X_\tau| \geq (1-\epsilon)\frac{n}{\Gamma}.$$

The proof compares the current distribution with those hypothetical local updates, then uses the entropy budget of the actual random-scan chain.

Step 1: entropy controls the single-vertex changes

Fix μ and write $a_v = d_{\mathrm{TV}}(\mu, \mu P_v)$.

Convexity and (H) give

$$D_{KG}(\mu) \geq \frac{1}{n} \sum_v \left(\Phi(\mu) - \Phi(\mu P_v) \right) \geq \frac{2}{n} \sum_v a_v^2.$$

Applying (CS) with $z_v = a_v^2$ gives

$$\frac{1}{n} \sum_v a_v \leq \sqrt{\frac{1}{n} \sum_v a_v^2} \leq \sqrt{\frac{D_{KG}(\mu)}{2}}.$$

Step 2: a sequence of small changes is still small



For two kernels A, B,

$$\mu AB - \mu = (\mu A - \mu)B + (\mu B - \mu).$$

The triangle inequality and contraction imply

$$d_{\mathrm{TV}}(\mu, \mu AB) \le d_{\mathrm{TV}}(\mu, \mu A) + d_{\mathrm{TV}}(\mu, \mu B).$$

- Both differences on the right are measured at the **original** distribution μ
- This matters: we only know the entropy bounds for updates starting from μ
- Repeat this argument for any finite sequence of kernels

The full telescoping calculation

For a sequence $P_{v_1}, \ldots, P_{v_\ell}$,

$$\mu P_{v_1} \cdots P_{v_\ell} - \mu = \sum_{i=1}^{\ell} (\mu P_{v_i} - \mu) P_{v_{i+1}} \cdots P_{v_\ell}.$$

The final suffix is the identity when $i = \ell$. Contraction and the triangle inequality give

$$d_{\mathrm{TV}}(\mu, \mu S_u) \leq \sum_v M_{uv} a_v,$$

$$d_{\mathrm{TV}}(\mu, \mu S_u P_u) \leq a_u + \sum_v M_{uv} a_v.$$

Step 3: move the local certificate back to μ

Use the ranges of x_u, y_u to bound the errors in (G):

$$1 \leq \beta \mathbb{E}_\mu x_u + \gamma \mathbb{E}_\mu y_u + \beta a_u + (\beta + \gamma d(u)) \sum_v M_{uv} a_v.$$

Sum over u, divide by n, and interchange the finite sums:

$$1 \leq \frac{\Gamma}{n} \mathbb{E}_\mu |X| + \frac{1}{n} \sum_v \left(\beta + \sum_u (\beta + \gamma d(u)) M_{uv} \right) a_v$$

$$\leq \frac{\Gamma}{n} \mathbb{E}_\mu |X| + C \frac{1}{n} \sum_v a_v.$$

The last step uses $\sum_u M_{uv} \leq \kappa$ and $\beta + \gamma d(u) \leq \Gamma$.

Step 4: spend the entropy budget

By step 1,

$$1 \leq \frac{\Gamma}{n}\mathbb{E}_{\mu}|X| + C\sqrt{\frac{D_{K^G}(\mu)}{2}}.$$

Average over $\mu_t = \delta_{\emptyset}(K^G)^t$ and use the same time-averaging calculation as for blocks:

$$1 \leq \frac{\Gamma}{n}\mathbb{E}|X_\tau| + C\sqrt{\frac{\Phi_0}{2T}}.$$

Rearranging proves the Glauber theorem.

Application 1: every graph

Recover the local occupancy parameters from lecture 08



Take $\beta = 1/p$, $\gamma = 1$, and let X_F be a hard-core random independent set of F. Then

$$\frac{1}{Z_F(\lambda)} + E_F(\lambda) = \mathbb{P}(X_F = \varnothing) + \mathbb{E}|X_F| \geq 1.$$

- Indeed, $|X_F| \geq \mathbf{1}_{\{X_F \neq \varnothing\}}$
- Thus every graph has these local occupancy parameters, with

$$\Gamma = \Delta + \frac{1}{p} = \Delta + 1 + \frac{1}{\lambda}$$

- The block theorem applies, with $2^{O(\Delta)}$ time per update

For Glauber, the empty list already works

Take S_u to be the identity kernel: do no neighborhood updates.

For every state I,

$$\beta(P_u x_u)(I) + \gamma y_u(I) = \mathbf{1}_{\{y_u(I)=0\}} + y_u(I)$$
$$\geq 1.$$

- If the neighborhood is empty, the first term is 1
- Otherwise the second term is at least 1
- Averaging proves (G) for every μ, with $\kappa = 0$ and $C = 1/p$

The general-graph guarantees

Starting from $\varnothing$, put $\Gamma = \Delta + 1/p$. For expected size at least $(1 - \epsilon)n/\Gamma$, the two theorems give:

Dynamics	Sufficient updates
Neighborhood block	$T \geq (\Gamma^2 + p^{-2})n \log(1 + \lambda)/(2\epsilon^2)$
Glauber	$T \geq n \log(1 + \lambda)/(2p^2\epsilon^2)$

- At $\lambda = 1$, Glauber gives expected size $(1 - \epsilon)n/(\Delta + 2)$ after $T \geq 2n \log 2/\epsilon^2$ updates
- For fixed finite λ, the denominator is $\Delta + 1 + 1/\lambda$; making λ large approaches the greedy bound

Application 2: triangle-free graphs

Why one sweep exactly resamples the neighborhood

Fix the exterior configuration, and list the vertices of $N(u)$ in any fixed order.

- Triangle-freeness means no two of these vertices are adjacent
- Whether a neighbor v is blocked depends only on the fixed exterior
- During a full sweep, each blocked vertex is forced to 0
- Each available vertex receives a fresh independent Bernoulli-p value
- This is exactly the conditional hard-core distribution on the available edgeless graph F, regardless of the initial interior

Therefore, as kernels on all independent sets, $S_u = Q_u$.

Both dynamics inherit every triangle-free certificate



- For one sweep at every center, a vertex v occurs once for each center in $N(v)$
- Thus $\kappa \le \Delta$ and we may use $C = \beta + \Delta\Gamma$
- Starting from $\emptyset$, every local occupancy pair from lecture 08 gives expected size $(1-\epsilon)n/\Gamma$ with

$$T_N \ge \frac{(\Gamma^2+\beta^2)n\log(1+\lambda)}{2\epsilon^2},$$

$$T_G \ge \frac{(\beta+\Delta\Gamma)^2 n\log(1+\lambda)}{2\epsilon^2}.$$

- These are block steps and single-vertex steps, respectively

Recall an explicit certificate with no special functions

For $\Delta \geq 2$, set

$$\lambda = 1, \quad t = \frac{1}{2}\log_2 \Delta, \quad \beta = 2^{t+1} = 2\sqrt{\Delta}, \quad \gamma = \frac{2}{t} = \frac{4}{\log_2 \Delta}.$$

For an edgeless F with y vertices, local occupancy asks for

$$\frac{\beta}{2^{y+1}} + \frac{\gamma y}{2} \geq 1.$$

- If $y \leq t$, the first term is at least 1
- If $y \geq t$, the second term is at least 1
- Hence $\Gamma = 2\sqrt{\Delta} + 4\Delta/\log_2 \Delta$

An explicit logarithmic improvement, now algorithmic

Using $\log_2 \Delta \leq 2\sqrt{\Delta}$ gives

$$\Gamma \leq \frac{8\Delta}{\log_2 \Delta}, \qquad \frac{n}{\Gamma} \geq \frac{n \log_2 \Delta}{8\Delta}.$$

- Either dynamics returns a set of expected size at least $(1 - \epsilon)n \log_2 \Delta/(8\Delta)$, using the run lengths below
- Here $\Phi_0 \leq n \log 2$, and all the update bounds are polynomial in $n, \Delta, 1/\epsilon$
- If $\Delta \leq 1$, take every isolated vertex and one endpoint from each isolated edge directly

Optional: transfer the optimized certificate too

For $\Delta \geq 1$, put $\ell = \log(1 + \lambda)$ and let $w > 0$ satisfy

$$we^w = \Delta\ell; \qquad w = W(\Delta\ell).$$

The optimized parameters from lecture 08 are

$$\beta = \frac{e^w}{p(1+w)}, \qquad \gamma = \frac{\ell}{p(1+w)}, \qquad \Gamma = \frac{\Delta\ell}{pw}.$$

To verify them, write $r = \ell y$. For any edgeless F with y vertices,

$$\beta p(1+\lambda)^{-y} + \gamma p y = \frac{e^{w-r} + r}{1+w} \geq 1,$$

using $e^x \geq 1 + x$ with $x = w - r$.

Optional: the leading constant 1

Let $\Delta \to \infty$ and choose $\lambda = 1/\log\Delta$.

- Then $\ell/p \to 1$ and $\Delta\ell \sim \Delta/\log\Delta$
- From $w + \log w = \log(\Delta\ell)$, we get $w/\log\Delta \to 1$
- Consequently

$$\Gamma = \frac{\ell}{p}\frac{\Delta}{w} = (1 + o(1))\frac{\Delta}{\log\Delta}$$

- Choose also $\epsilon = 1/\log\Delta$
- Both dynamics then return a set of expected size at least $(1 - o(1))n\log\Delta/\Delta$; the explicit bounds below remain polynomial

Algorithms with an expected-size guarantee

A random time is an analysis device



- Our theorems concern X_τ, for an independent uniform time $\tau < T$
- They do not claim the same bound for the final state X_T
- Let Y be the **size** of the largest set visited in a run, including times $0, \ldots, T$
- For every trajectory and every $\tau < T$, we have $Y \geq |X_\tau|$
- Thus $\mathbb{E}Y \geq \mathbb{E}|X_\tau|$, and the algorithm need not choose τ or compute its current distribution

Explicit choices for the complete algorithm

Fix $0 < \epsilon < 1$, start from $X_0 = \emptyset$, and use the budget $\Phi_0 \leq n \log(1 + \lambda)$. Run the chosen dynamics for

$$T_N = \left\lceil \frac{(\Gamma^2 + \beta^2) n \log(1 + \lambda)}{2\epsilon^2} \right\rceil,$$

$$T_G = \left\lceil \frac{C^2 n \log(1 + \lambda)}{2\epsilon^2} \right\rceil.$$

- General graphs: $\Gamma = \Delta + 1/p$, and for Glauber take $C = 1/p$
- Triangle-free graphs: use either certificate above and $C = \beta + \Delta\Gamma$
- Return the largest set visited, X^{alg}. Then $\mathbb{E}|X^{\text{alg}}| \geq (1 - \epsilon)n/\Gamma$

Update cost versus number of updates



- With adjacency lists, a Glauber update costs $O(\Delta + 1)$ operations and random draws
- A general block update enumerates subsets in $2^{O(\Delta)}$ time
- A triangle-free block can check availability in $O((\Delta + 1)^2)$ time, then toss at most Δ coins
- Multiply by T_N or T_G to bound update work
- Copying the current set whenever it is best adds at most $O(n)$ per step, preserving the polynomial bounds for Glauber and triangle-free blocks

What we proved

Graph class	Lower bound on expected output size	Dynamics
Every graph	$(1-\epsilon)n/(\Delta+1+1/\lambda)$	Glauber
Triangle-free, $\Delta \geq 2$	$(1-\epsilon)n\log_2\Delta/(8\Delta)$	Both
Triangle-free, $\Delta \to \infty$	$(1-o(1))n\log\Delta/\Delta$	Both

- The general block theorem is valid too, with exponential dependence on Δ in its update cost
- Glauber is polynomial-time for fixed $\lambda > 0$; both triangle-free choices above also give polynomial bounds, including the specified varying λ

Reading

- Lecture 08 and [DaKa25] for local occupancy and hard-core conditioning
- [DJPR17] for the triangle-free occupancy bound
- Dynamics proofs: Ewan Davies and Brian Tan, *Large independent sets via neighborhood block dynamics, Glauber dynamics, and local occupancy* (manuscript), the neighborhood block theorem, Glauber transfer theorem, and one-sweep argument
- We used the exact-certificate case of the transfer theorem; approximate certificates and more general neighborhood structures are further applications

References

[DaKa25] Davies, E., Kang, R. J. (2025). *The hard-core model in graph theory*. URL: https://arxiv.org/html/2501.03379v2

[DJPR17] Davies, E., Jenssen, M., Perkins, W., Roberts, B. (2017). *On the average size of independent sets in triangle-free graphs*. Proceedings of the American Mathematical Society, 146 (1), 111–124. DOI: https://doi.org/10.1090/proc/13728